

The model that wins on cross-validation loses on a new machine: extrapolation, boundedness and physics constraints in confinement scaling

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31 August 2026

Abstract

Energy confinement time sets the size a tokamak must reach to ignite, and the field predicts it with power-law scaling laws fitted across existing devices. On the ITPA global H-mode database (HDB5, standard analysis set STD5; 6228 quasi-stationary slices from 4471 discharges across 18 tokamaks) a random forest beats the published IPB98(y,2) scaling law by 36% in RMSLE under grouped cross-validation by discharge. That margin does not survive the question a scaling law exists to answer. Under leave-one-tokamak-out the ordering of the three primary fitted models is exactly reversed: the random forest is worse than a log-linear power law on 13 of 13 held-out machines, paired 95% interval $[+0.157, +0.342]$. The trees' errors track the distance of the held-out machine from the training data ($\rho = +0.85$) where the power law's do not ($\rho = -0.06$), and a random forest cannot predict outside its training target range: on the largest device 48% of rows lie above anything it can output. A size-ordered cut inside the database reproduces ITER's $1.82\times$ jump; there the tree ensembles score closer to a constant predictor than to the power law, and split-conformal intervals calibrated to 90% cover 3% of rows at unchanged width: the models are not vague out of distribution, they are confidently wrong. Two repairs work, and both impose structure on long-range behaviour rather than adding flexibility. The Connor–Taylor dimensional constraints, imposed as linear equality constraints on the exponents, give 0.183 at that cut, the best blind score here, with nothing to tune. And a Gaussian process whose kernel is flexible but unbounded gives the best cross-validated score here, 0.112, with 0.191 at the cut and 92.5% coverage of a never-recalibrated nominal 90% interval, where the same process with a bounded kernel scores 1.948, worse than a constant. What fails out of distribution is therefore not flexibility but boundedness, correcting this paper's own first diagnosis. The reversal reproduces on rows the standard set does not contain, in a second confinement regime, and on two biological scaling laws. For ITER the surviving and failing models differ by a factor of 8.3, which we record as a locked prediction.

1 Introduction

A tokamak ignites when the alpha heating from its own fusion reactions covers everything the plasma loses. The energy confinement time τ_E sets that loss rate, which makes it the quantity that decides how large a next-step device has to be, and no first-principles calculation predicts it to the accuracy a design decision needs. The field's working substitute is an empirical scaling law: a power law in the engineering parameters,

$$\tau_E = C I_p^{a_1} B_t^{a_2} \bar{n}_e^{a_3} P^{a_4} R^{a_5} \epsilon^{a_6} \kappa^{a_7} M^{a_8}, \quad (1)$$

fitted by least squares across the discharges of every machine that has run. In log space Eq. (1) is exactly a linear model, so fitting one is ordinary least squares on a log design matrix. The reference fit is IPB98(y,2) [1], a regression over the ITPA multi-machine database whose exponents were used to size ITER; that database is still maintained and republished [2].

What such a law is used for is an extrapolation, and a large one. ITER's major radius is 1.82 times the largest machine in the database, and SPARC [10] is designed for a toroidal field three times anything in it. The law is asked for a number at a point no row occupies, and that is the regime in which a fitted regression carries no guarantee at all.

Replacing Eq. (1) with a flexible regressor is an obvious move, and machine learning has been productive elsewhere in this field: disruption prediction [12], transfer of a disruption predictor to a device that supplied almost none of its training data [13], and real-time magnetic control [14]. Confinement scaling itself has

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been revisited with symbolic regression, which abandons the power-law form rather than fitting it [15], and with a neural residual learned on top of a frozen power law [16]. Measured conventionally the flexible models win, and on the rows used here the margin over the published law is 36%.

That measurement is the wrong one, for a reason about the data rather than about the models. Cross-validation on this database holds out *discharges*, and every machine in a held-out fold also appears in the training fold, so a held-out row has hundreds of near neighbours recorded on the same device at nearby settings. What the score quantifies is how much of JET is predictable from the rest of JET. That a validation split should match the generalisation being claimed, and that grouped observations flatter a model when it does not, is well understood in fields whose data arrives in clusters [17]; it is not how a candidate confinement model is usually scored. A scaling law is not for the machines that exist.

This paper measures what the honest split costs. We escalate what is held out in three steps, a discharge, then a whole machine, then an entire size range, and score the same models under each. The results are: (i) under leave-one-tokamak-out the ranking of the three primary fitted models is the exact reverse of their ranking under grouped cross-validation, and the best cross-validated model loses to a log-linear power law on 13 of 13 machines; (ii) three measurements of why, covering extrapolation distance, the training-target bound of a random forest, and what polynomial flexibility costs in the tail; (iii) a size-ordered split inside the database that reproduces ITER’s extrapolation factor, at which conformal intervals calibrated to 90% cover 3% of rows without becoming any wider; (iv) the Connor–Taylor dimensional constraints, imposed as linear equality constraints on the exponents, beating every model fitted here at that cut with nothing to tune; (v) a Gaussian-process experiment that separates flexibility from boundedness and corrects the diagnosis the earlier sections support; (vi) replication on rows the standard analysis set does not contain, in a second confinement regime, and on two scaling laws from biology, with a predictor-count ladder that separates the extrapolation failure, which persists at every predictor count tested, from the ranking inversion, which does not appear below three; and (vii) a locked forecast for SPARC, JT-60SA and ITER. Section 3 begins with a property of the design matrix itself: it is rank deficient by two, and one of the two dependencies is that a published scaling law added as a model feature contributes nothing a log-linear fit can use.

2 Data and method

Data. The ITPA Global H-mode Confinement Database, standard analysis set STD5 v5.2.3 [2], obtained from OSF. After requiring finiteness and positivity, 6228 rows from 4471 discharges across 18 tokamaks remain. The target is the thermal confinement time τ_{AUTH} . No synthetic data appears in any result. The analysed file is pinned by SHA-256 and verified on load, so every number below refers to one specific set of bytes rather than to whatever the host currently serves.

Features and models. Nine log engineering features: plasma current, toroidal field, line-averaged density, loss power, major radius, elongation, inverse aspect ratio, effective mass and minor radius. The analytic IPB98 prior is *excluded* as a feature throughout the extrapolation arms, because its exponents were fitted on this same database including whichever machine is held out; retaining it leaks the answer into every fold. The zoo is a ridge regression on the log features (i.e. a fitted power law), a random forest [20] (300 trees), a histogram gradient booster, polynomial ridge controls of degree 2 and 3, and a mean baseline. IPB98(y,2) evaluated analytically is reported as a physics reference, and is marked throughout as *not blind*.

Splits. (i) Grouped cross-validation by discharge. (ii) Leave-one-tokamak-out over the 13 machines with at least 30 rows. (iii) A size-ordered cut: machines are ordered by median major radius, and the model is trained below a cut and scored on every machine above it. Scores are the root-mean-square error in $\log \tau$, $\sqrt{(\log \hat{\tau} - \log \tau)^2}$, abbreviated RMSLE here and in the figures. This is the plain log residual rather than the $\log(1 + y)$ variant the same acronym sometimes denotes; confinement times are strictly positive, so no offset is needed. Intervals are percentile bootstraps resampling whole *machines*, since the claim concerns behaviour on an unseen machine.

3 The design matrix is rank deficient by two

Standardised, the ten-feature matrix has rank 8, with the last two singular values at 4.9×10^{-14} and 2.4×10^{-14} . Projecting candidate vectors onto the null space confirms two exact dependencies at relative

Table 1: The same models and the same nine blind features under two splits. Only the held-out unit changes. IPB98 is a reference, not a blind competitor.

Model	CV	LOMO	Ratio	CV rank	LOMO rank
random forest	0.128	0.465	$3.6\times$	1	5
hist gradient boosting	0.130	0.359	$2.8\times$	2	4
ridge, log-cubic	0.148	0.849	$5.7\times$	3	6
ridge, log-quadratic	0.158	0.300	$1.9\times$	4	3
ridge, log-linear	0.181	0.214	$1.2\times$	5	2
IPB98(y,2), analytic	0.199	0.188	$1.0\times$	6	1
mean baseline	0.869	0.994	$1.1\times$	7	7

residuals of order 10^{-16} : minor radius is derived as $a = \epsilon R$ during cleaning, and the IPB98 prior is a fixed log-linear combination of the other eight features.

The second is the substantive one. Adding a published physics scaling as a model feature feels like adding knowledge; in log space it provably is not, and it contributes exactly nothing to a log-linear fit. Nothing failed loudly because `scipy.linalg.lstsq` inverts through the SVD pseudoinverse and silently returns the minimum-norm member of a two-parameter family of coefficient vectors that fit identically well.

We fell into two methodological traps first. The null space is a *subspace*, so when the deficiency exceeds one the basis vectors printed by the SVD generally do not resemble the dependency sought even when it is exactly right; the correct test is $\|v - N^T N v\|$. And standardisation rescales the null space, so a raw dependency must be mapped into standardised coordinates before comparison. Omitting this gives residuals of 0.41 and 0.96 for dependencies that are exactly true.

4 Refitting, and where the disagreement lives

Refitting Eq. (1) on all STD5 rows (design matrix 6228×9 , condition number 10.7) gives I_p 1.08 against the published 0.93 and R 1.58 against 1.97, while P and B_t return almost exactly. This disagreement is expected: IPB98 was fitted to a selected ITER-relevant subset, whereas this fit applies no selection criteria and includes the spherical tokamaks. A refit on a different population is a different regression.

The normal equations, QR and the SVD pseudoinverse were each implemented from scratch, including the triangular substitutions, and agree to 7.6×10^{-13} . Cholesky on the normal equations is five times faster and three orders of magnitude less accurate, which is the expected trade: it forms $X^T X$ and so works at the *square* of the condition number. At $\kappa = 10.7$ that costs nothing real, and it is the only one of the three that fails loudly on a rank-deficient matrix.

Where the two differ matters more than by how much. Decomposing the difference between the exponent vectors along the singular directions of the design matrix, 77% of it lies in the single weakest direction, which carries 0.3% of the matrix’s variance, while the three strongest directions carry 82% of the variance and account for 0.75% of the disagreement. That weak direction is plasma current traded against machine size, which is weak for a structural reason: tokamaks are not designed to vary the two independently, so no number of additional shots from existing machines separates them.

5 The ranking inversion

Table 1 is the central result. Among the three primary models, a power law, a random forest and a gradient booster, the order under one split is the exact reverse of the order under the other. The random forest is the best model in the study by cross-validation and the worst of the three on a machine it has not seen. The two polynomial rungs are controls, scored in the same table but outside the ranking claim, and including them breaks the inversion rather than extending it: the log-cubic rung is third of the five blind models under cross-validation and last of them on a held-out machine.

Because the machines differ enormously in difficulty, the marginal bootstrap intervals overlap and are the wrong test; resampling the *paired* per-machine difference removes that shared difficulty. The random forest is worse than the log-linear power law on **13 of 13** machines, mean gap +0.251, 95% interval [+0.157, +0.342]. The gradient booster is worse on 12 of 13.

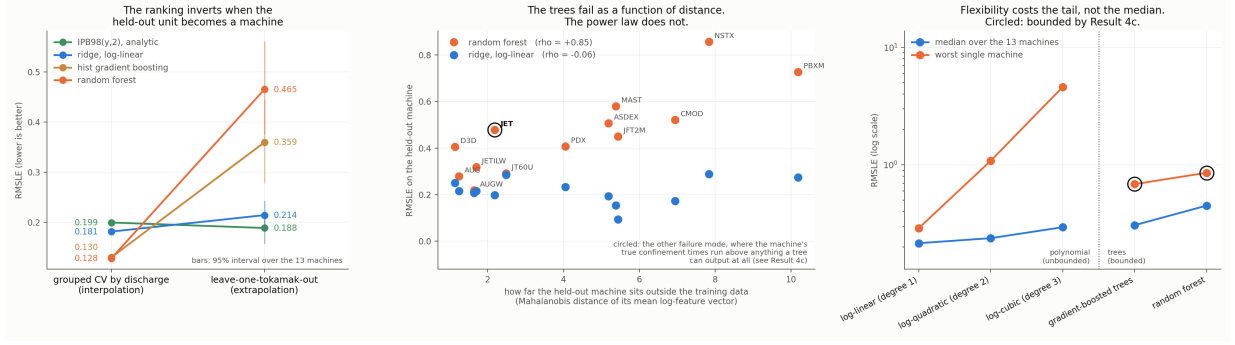


Figure 1: Interpolation against extrapolation. Left: each model under both splits; the lines crossing is the result. Middle: per-machine error against Mahalanobis distance from the training data, rising for the trees and flat for the power law. Right: the flexibility ladder, where what grows with flexibility is the tail rather than the centre.

Table 2: Per-machine RMSLE under leave-one-tokamak-out, ordered by Mahalanobis distance of the held-out machine’s mean log-feature vector from the training mean (computed through the pseudo-inverse, since that covariance is singular by Sec. 3). The trees degrade with distance; the power law does not.

Held out	rows	dist.	IPB98	ridge	hist GB	forest
D3D	388	1.1	0.252	0.251	0.246	0.406
AUG	1377	1.2	0.197	0.216	0.281	0.279
AUGW	767	1.6	0.218	0.207	0.212	0.219
JETILW	866	1.7	0.257	0.216	0.244	0.318
JET	1762	2.2	0.148	0.198	0.487	0.478
JT60U	100	2.5	0.275	0.285	0.307	0.291
PDX	97	4.0	0.227	0.233	0.323	0.407
ASDEX	431	5.2	0.131	0.194	0.207	0.507
MAST	39	5.4	0.136	0.154	0.317	0.581
JFT2M	69	5.4	0.095	0.094	0.234	0.450
CMOD	45	6.9	0.119	0.173	0.569	0.521
NSTX	185	7.8	0.222	0.289	0.686	0.857
PBXM	59	10.2	0.172	0.274	0.559	0.727

5.1 Three mechanisms

Distance. Ranking each model’s per-machine error against the Mahalanobis distance of that machine’s mean log-feature vector from the training mean gives $\rho = +0.85$ for the random forest and $+0.54$ for the gradient booster, against $\rho = -0.06$ for the log-linear power law. The trees’ errors are explained by extrapolation distance. The power law’s are not: its error is uncorrelated with how far the machine sits from anything it saw.

On the four most distant machines (PBX-M, NSTX, C-Mod, JFT-2M, all small, one of them spherical) the forest’s error is 2.7 to 4.8 $\times$ the power law’s; on the four nearest it is within 1.7 $\times$.

A hard bound. A random forest predicts by averaging leaf values, and every leaf value is itself a mean of training targets, so every prediction it can make lies in $[\min y_{\text{train}}, \max y_{\text{train}}]$ whatever the features say. With JET held out, 48% of its rows exceed the maximum confinement time in the remaining 12 machines and its best shot is 3.7 $\times$ above anything any tree in the forest can output. This is not a shortfall more data would close; it is the functional form.

A gradient booster carries no such guarantee. It sums tree outputs onto an initial estimate rather than averaging targets, and nothing in that construction confines the sum to the training range, so the argument above does not transfer to it. What we can report is that it never uses the freedom: over the 13 leave-one-out folds and the size cut of Sec. 6, its largest prediction stays below $\max y_{\text{train}}$ on every one, coming closest with JET held out at 0.066 in logs beneath the ceiling and sitting 0.567 beneath it across the size cut. Its boundedness on these rows is measured rather than guaranteed, and every claim below that needs the guarantee is made about the forest.

Flexibility costs the tail, not the centre. A polynomial ladder in the log features (degrees 1–3) is flexible but still extrapolates. Degree 2 is *not* meaningfully worse than degree 1 on a typical machine:

Table 3: Three questions of increasing difficulty. *Skill* places each model between the mean baseline (0.00) and the analytic power law (1.00).

Model	shot	machine	larger machine	skill
IPB98(y,2)	0.199	0.188	0.194	1.00
ridge, log-linear	0.181	0.214	0.278	0.93
random forest	0.128	0.465	0.938	0.41
hist grad. boosting	0.130	0.359	1.072	0.31
mean baseline	0.869	0.994	1.459	0.00

median 0.238 against 0.216, better on 8 of 13, paired interval $[-0.013, +0.237]$ containing zero. What flexibility costs is the tail. Degree 1’s worst machine is 0.289; degree 2’s is 1.083 and degree 3’s is 4.601, both on C-Mod. Boundedness is simultaneously why neither ensemble extrapolates and why neither blows up: their worst cases, 0.686 and 0.857, sit far below degree 3’s.

6 The size extrapolation ITER asks for

Holding out JET still leaves twelve machines spanning much of its range, so leave-one-out extrapolates in identity while interpolating in size. ITER’s major radius is 6.2 m against 3.40 m for the largest row here, a factor of 1.82. That factor is available inside the database: training on the 14 smallest machines (up to DIII-D, 1.865 m, 3498 rows) and predicting the 4 largest (up to JT-60U, 3.40 m, 2730 rows) demands a ratio of 1.823 against ITER’s 1.824. The cut is selected by proximity to that ratio rather than by eye, so it is a property of the data.

The power law retains 93% of the distance from a constant predictor to the analytic law; the trees retain 41% and 31%. The gradient booster scores 1.072 where predicting a single constant scores 1.459. Asked the question a scaling law exists to answer, the best cross-validated models land closer to a constant than to the law they beat by 36%. The ordering is identical on all three individually scored held-out machines, so the pooled number is not an artifact of JET’s weight, and dropping the spherical tokamaks moves the random forest from 0.938 to 0.936, so the effect is size rather than plasma shape.

7 A power law with a bounded correction

The diagnosis implies a cure. Fit the power law, learn a correction on its log residuals, and damp it by λ :

$$\log \tau = \underbrace{f_{\text{ridge}}(x)}_{\text{unbounded, log-linear}} + \lambda \underbrace{g(x)}_{\text{correction on residuals}}. \quad (2)$$

At $\lambda = 0$ this is exactly the ridge row of Table 1. Anchoring a learned correction on a frozen power law is also the design used in concurrent work on this database [16]; what is varied here is the one property the diagnosis points at. Two correction families differ in precisely the property that matters: a degree-2 log polynomial, which is unbounded, and depth-2 boosted trees, which as measured below never leave the residual range they were trained on.

Across the ITER-matched cut the two go opposite ways. Both start at 0.278; the polynomial correction climbs to 0.356 and the bounded correction falls monotonically to **0.206**, better than plain ridge on all three held-out machines and by the widest margin on JT-60U, the most distant (0.440 to 0.281). It is the best blind model at that cut, 26% below plain ridge and $4.6\times$ below the random forest, and within 6% of IPB98, which was fitted with those machines included.

The mechanism is measurable. Fitted on the 14 small machines, the base power law is *biased* on the held-out ones: mean log residual -0.218 , so it over-predicts by about 20%, while its scatter (0.172) is no worse than in-sample (0.187). The tree correction never leaves the range it was trained on, on any held-out row, and inside that bound it points the right way and is largest where the bias is largest. The same boundedness that makes these ensembles useless as predictors of τ on a larger machine makes them safe as correctors, because the quantity they are bounded on is no longer a target that grows with machine size but a residual centred on zero.

Two honest limits. Cross-validation selects $\lambda = 1$ in both families, while the rung best on a held-out machine is $\lambda = 0$; nothing in the CV signal points at it. And scored at every rung of the size sweep rather than only the matched one, the bounded hybrid wins at 5 of 8 well-powered cuts, not all, with no rule over

Table 4: Empirical coverage of nominal 90% split-conformal intervals, and the multiplicative half-width under leave-one-out.

Model	CV	LOMO	ITER cut	width
IPB98(y,2)	90%	89%	88%	1.40×
ridge, log-linear	90%	83%	70%	1.33×
bounded hybrid	90%	64%	76%	1.26×
hist grad. boosting	91%	45%	0%	1.23×
random forest	91%	35%	3%	1.23×

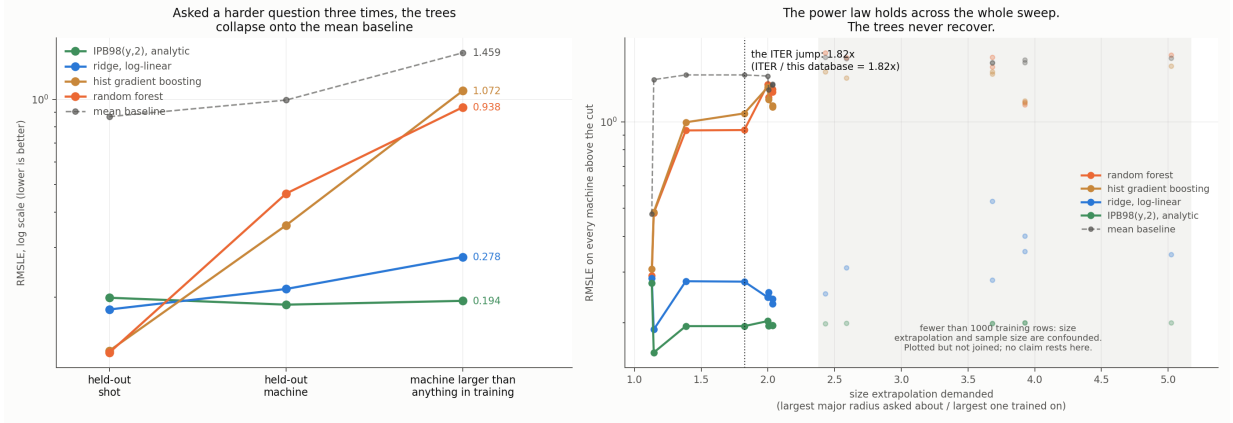


Figure 2: Size-ordered extrapolation. Left: the escalation across three splits. Middle: the per-machine breakdown at the ITER-matched cut. Right: the sweep across every cut, with underpowered cuts marked and excluded from claims.

those eight points separating wins from losses. The claim that survives is narrow: at the cut matching the jump to ITER, and robustly across all nine correction settings tried there, a bounded correction beats the power law and the reason is understood.

8 Calibrated intervals and their collapse

For a next-step device the point error is not the deliverable; the interval is. We calibrate split-conformal intervals [3] on log residuals, holding back 25% of *discharges* to calibrate, which gives finite-sample coverage under exchangeability of the calibration and test rows.

That proviso is the result. Exchangeability holds under grouped CV, where every model lands within a point of nominal, and that control is what licenses reading the rest of Table 4. It fails by construction when the test rows come from a machine the model never saw; that exchangeability is the binding assumption, and that repairing it requires knowing the shift, is the subject of the weighted-conformal literature [18], and calibrated uncertainty degrading under dataset shift is a general finding rather than a fusion one [19]. The random forest’s 90% interval then covers 35% of rows on an unseen machine and 3% across the ITER-matched cut; the gradient booster covers none of the 2730 rows at all.

The widths barely move: no model’s interval changes width by more than 1.5% between the two arms, because the half-width is set by calibration rows drawn the same way in both. The intervals do not become vague out of distribution. They stay the same size and miss.

Coverage collapses along the same axis the errors grow, but only for the trees: $\rho = -0.77$ for the random forest and -0.54 for the gradient booster against $+0.27$ for the power law, mirroring the point-error correlations of Sec. 5.1 with the sign flipped. The power law’s intervals are somewhat too narrow everywhere rather than catastrophically too narrow far away, and for a next-step device that distinction is most of what matters, because distance is the one thing known in advance. None of this is a defect in conformal prediction. It is a measurement of an assumption being false.

Table 5: Fitting under the Connor–Taylor constraint hierarchy. The only difference between the first and last rows is the constraint: same features, same solver, no hyperparameter.

Model	in sample	CV	LOMO	ITER cut
power law, collisionless	0.1818	0.182	0.206	0.183
IPB98(y,2), analytic*	—	0.199	0.188	0.194
bounded hybrid (Sec. 7)	—	0.151	0.246	0.206
power law, electrostatic	0.2245	0.225	0.241	0.237
power law, Kadomtsev	0.1808	0.181	0.210	0.254
power law, unconstrained	0.1808	0.181	0.214	0.279

*fitted on this database including the held-out machines; not a blind model.

9 One line of dimensional analysis

Every model to this point learns its functional form from the data. None is told any physics. That is worth revisiting, because the field has known since Connor and Taylor [5] that a confinement scaling law is not free: requiring it to be expressible in the dimensionless plasma variables imposes *linear equality constraints on the exponents*. A physics assumption is therefore a matrix C , and the fit becomes

$$\min_b \|Xb - y\|^2 \quad \text{subject to} \quad Cb = d, \quad (3)$$

which the KKT system solves in closed form. Nothing new is needed to run the experiment: the solver is the one written by hand for Sec. 4.

We derive the constraints in code from the definitions of ρ^* , β and ν^* rather than transcribing exponents from a paper, which makes the derivation checkable against something external. IPB98(y,2) was published in 1999 and was never told about these surfaces. It lands on the Kadomtsev surface [11] at a distance of 0.00096 and on the collisionless surface at 0.0045, both inside the rounding of its own two-decimal exponents. A transcription error would not reproduce that.

At the ITER-matched cut the collisionless power law scores **0.183**, the best of any blind model in this study (Table 5). It beats the bounded hybrid of Sec. 7, which was built specifically for that cut, and it beats the analytic law that was fitted with those machines included. It has no hyperparameter and nothing to tune.

The two constraints behave differently. The Kadomtsev constraint is *free*: at 0.1808 in sample it is identical to the unconstrained fit to four decimals, because the data already satisfies it unaided. It nonetheless beats the unconstrained fit at 15 of 15 cuts in the size sweep, since a constraint the data satisfies on average still stops the fit wandering when the training half is small. The collisionless constraint costs 0.001 in sample and wins at 8 of 8 well-powered cuts. Push one rung further to a beta-independent law and it degrades sharply, to 0.237. There is an optimum in the middle of the hierarchy, and nothing in this analysis predicted where it would fall; as with the hybrid, cross-validation does not select it, because every constrained fit is slightly worse than unconstrained under CV by construction.

It matters that this is a constraint and not a prior. Supplying the same physics as a Gaussian prior instead, shrunk along the weak singular direction of Sec. 4, is worth much less. Targeting the shrinkage beats isotropic shrinkage at every matched penalty strength, which is a real effect, but nothing in that family beats simply adopting IPB98’s published exponents outright. A constraint names a surface; a prior names a point. The weaker information is worth more, because the surface is what the physics actually asserts.

10 Repairing the intervals, and the limit of the repair

The diagnosis in Sec. 8 makes a falsifiable prediction. If the interval collapse is really about exchangeability rather than about the models, then calibrating on held-out *machines* instead of held-out discharges should repair it, and should repair it only as far as that unit reaches. We test both halves.

Both halves land. On a held-out machine the repair is essentially complete: every model returns to within two points of nominal, tree ensembles included, and the random forest goes from 35% to 88%. Across the size cut none does. No recalibration makes the two exchangeable there, because every calibration machine is smaller than every test machine, and scaling the interval by extrapolation distance recovers only part of the gap (3% to 40% for the forest). A repair that stops exactly where the diagnosis says it should is stronger evidence for the diagnosis than one that worked everywhere would be.

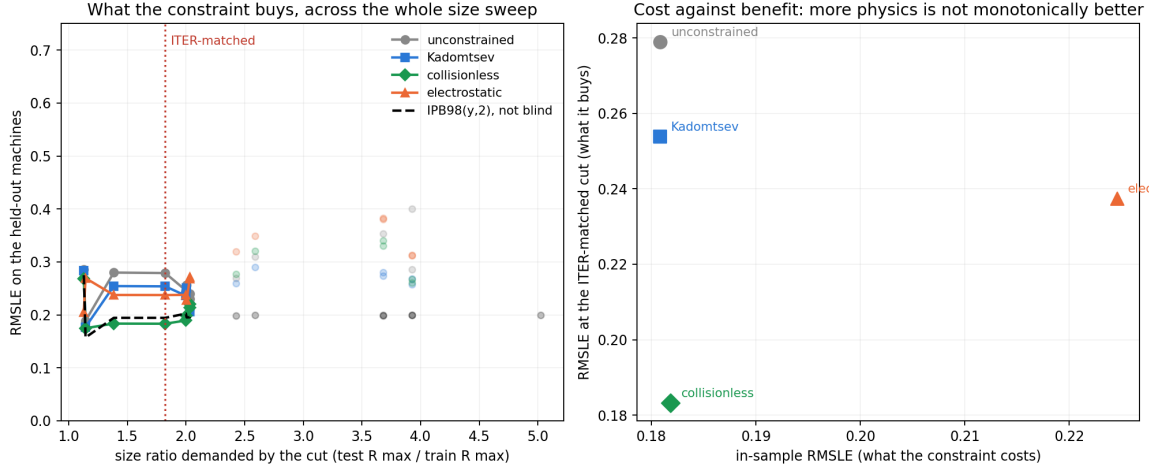


Figure 3: Cost against benefit across the size sweep. Left: the in-sample penalty each constraint imposes. Middle and right: the score at every size cut for the constrained and unconstrained fits, with underpowered cuts marked. The constraint costs almost nothing where the data is plentiful and pays where it is not.

Table 6: Empirical coverage of nominal 90% intervals under three calibration schemes, on a held-out machine and across the ITER-matched size cut.

Model	split conformal		by machine		+ distance	
	LOMO	cut	LOMO	cut	LOMO	cut
random forest	35%	3%	88%	29%	88%	40%
hist grad. boosting	45%	0%	90%	4%	89%	13%
ridge, log-linear	83%	70%	91%	77%	92%	82%
power law, collisionless	85%	91%	91%	95%	92%	92%
IPB98(y,2)*	89%	88%	89%	89%	89%	87%

The constrained power law of Sec. 9 is the single exception in Table 6: its intervals hold at the ITER-matched cut under every scheme, including plain split conformal, which fails there for everything else.

11 Replication on rows the standard set does not contain

Everything above rests on one file, which is the honest ceiling on all of it. The same OSF project publishes the full database revision, DB5.2.3, with 14153 rows against STD5’s 6228, and we pin it by its own SHA-256. Matching on (tokamak, shot, time) shows STD5 is an ELMy-H-mode quality selection out of it and leaves two populations this study had not analysed: **5358 H-mode rows STD5 does not contain** (12 machines, zero row overlap) and **3860 ohmic and L-mode rows** in a different confinement regime, which we score against ITER89-P [6] because an H-mode law is the wrong baseline for an L-mode plasma.

This replication has one way to be silently fake. If the row match against STD5 fails, every STD5 row stays in the "disjoint" arm, the arm becomes a superset of the data Sec. 5 was computed on, and it reproduces Sec. 5 by construction. We therefore check the matcher positively as well as negatively: STD5 matched against itself must overlap completely, and the disjoint arm must share zero rows.

The ranking inverts in both arms. On the disjoint H-mode rows the best cross-validated model beats the published law by 42%, a margin of the same size as the 36% this paper opens with and computed on rows that headline never saw, and the best model on an unseen machine is IPB98(y,2). On the ohmic and L-mode rows the cross-validation gain over ITER89-P is 67% and the best model on an unseen machine is the collisionless power law of Sec. 9. Counting machine–model pairs where a tree ensemble loses to the published law on an unseen machine: 19 of 24 and 10 of 10.

So the reversal is not an artifact of the standard set’s selection criteria, and not a property of ELMy H-mode or of IPB98(y,2) specifically. It is emphatically *not* an independent database: both arms come from the same ITPA collection and the same devices, and the five-machine arm is too small to carry a claim alone.

Table 7: Predicted energy confinement time, with nominal 90% intervals. Locked 31 August 2026 against a fixed dataset digest.

	SPARC	JT-60SA	ITER
	1.85 m	2.96 m	6.2 m
Model	(operating)		
IPB98(y,2)*	0.765 s	0.479 s	3.591 s
power law, collisionless	0.724 s	0.428 s	2.837 s
ridge, log-linear	0.720 s	0.444 s	2.858 s
random forest	0.136 s	0.449 s	0.435 s
hist grad. boosting	0.305 s	0.418 s	0.444 s

12 A locked prediction for three machines that have no data

Everything above is retrospective. To make the disagreement falsifiable rather than merely described, we record what each model predicts for three real devices, with intervals, a date, and a digest over the input rows so that a later edit leaves a mark. The parameter sets are published design values, and each reproduces the confinement time its own source quotes: SPARC to 0.6% and ITER to 2.9%.

Table 7 is the argument of this paper in one place. On JT-60SA, which sits inside the database’s size range and is already operating, all five models agree to within 15%. On ITER, $1.82\times$ beyond it, they disagree by a factor of **8.3**.

That gap cannot be closed by tuning, and Sec. 5 says why: the random forest predicts by averaging training targets, so its output cannot exceed the largest one, which here is 1.321 s. The gradient booster is not bound by that arithmetic and lands in the same place anyway. The random forest that beats the published law by 36% under cross-validation is arithmetically incapable of returning ITER’s predicted confinement time whatever it is asked, and its nominal 90% interval at ITER runs from 0.19 s to 0.98 s, containing neither the physics answer nor anything near it.

The JT-60SA column becomes checkable first, but not as of September 2026: the first campaign ran L-mode divertor plasmas, and its published power balance reports confinement following L-mode scaling, whereas the row above is an H-mode prediction scored against IPB98(y,2). H-mode operation is planned for the second campaign, beginning in the second half of 2026. The prediction fails if a stationary JT-60SA H-mode near this design point reports a thermal confinement time outside 0.349 s to 0.657 s. All five models place their intervals across roughly that range, so such a measurement would fail all of them together and one inside separates none of them: this row tests whether any model is right, not which, and that is why the ITER column carries the argument.

Nothing above anticipated one feature of this table: SPARC sits further from the training data than ITER does, at a Mahalanobis distance of 6.9 against 4.7, despite being the smaller machine. Its 12.2 T field is far outside a database that tops out near 4 T. Size is not the only direction in which a next-step device leaves the data behind, and an extrapolation audit organised around size alone would have missed it.

13 The same audit on a scaling law from another science

Every result to this point rests on ITPA data, so the last question is whether any of it is about fusion. Mammalian basal metabolic rate against body mass is the same object and the older one: Kleiber [4] found in 1932 that metabolic rate scales as mass to the $3/4$, and that exponent is still the published baseline a fitted model has to beat. Taxonomic *order* is the unit a species belongs to, as a tokamak is the unit a discharge belongs to, and *body mass* is the axis a next instance lies beyond, as machine size is. The data is Supplement 1 of a metabolic-rate compilation deposited on Figshare (article 3549807), pinned by SHA-256 as the HDB5 files are, giving 541 species records across 11 orders whose median masses span $342\times$. The analysis runs through the domain-agnostic module released with this work rather than a copy of the fusion pipeline, so it is the procedure a reader would apply to their own data, and the problem has a *single predictor*: no model can win by finding a better combination of features, so the only thing separating them is functional form. Refitting the exponent freely gives 0.687 against Kleiber’s 0.75, so the constraint is not free here either.

The extrapolation failure reproduces. Both tree ensembles are worse than both power laws at all 8 mass cuts, and the random forest loses to Kleiber on 9 of 11 held-out orders. The mechanism reproduces

Table 8: Kleiber’s law under the same three splits. Scores are RMSLE.

Model	CV, by species	leave-one-order-out	widest mass cut
Kleiber, exponent 3/4	0.396	0.440	0.374
power law, free exponent	0.374	0.432	0.496
hist grad. boosting	0.418	0.487	0.889
random forest	0.437	0.517	0.710

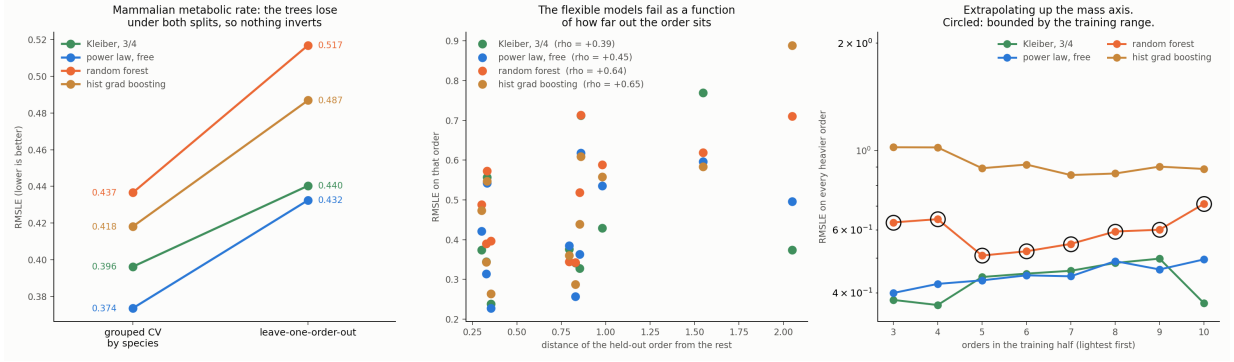


Figure 4: Kleiber’s law under the same three splits. Left: no inversion, because the trees lose under both splits. Middle: error against extrapolation distance. Right: the mass-ordered sweep, with the random forest’s training-range-bounded cuts circled.

with it: per-order error tracks extrapolation distance at $\rho = +0.64$ and $+0.65$ for the trees against $+0.39$ and $+0.45$ for the power laws, a narrower contrast than Sec. 5.1’s but the same sign, and the training-range bound binds at every cut.

The ranking reversal does not, and that bounds the claim. The trees never win the easy split either: under cross-validation by species they score 0.437 and 0.418 against the free power law’s 0.374. With no inflated margin there is nothing available to invert. The 36% gain on HDB5 came from a flexible model exploiting structure across nine features; with one predictor and a relationship close to a straight line in logs, a tree has far less to exploit and does not buy the interpolation win that later reverses. The reversal requires enough feature dimensionality for the flexible model to win interpolation first, which no single database could have shown.

The constraint result reproduces in direction, not in strength. Kleiber’s published exponent costs $+0.023$ under cross-validation and wins the widest cut, 0.374 against 0.496, where the held-out order is 69x heavier than the training median: Sec. 9’s trade exactly. But it is weaker, winning at 4 of 8 cuts rather than 8 of 8, decisively at the extremes and narrowly losing in the middle. That a constraint helps out of distribution replicates; that it helps reliably does not.

14 The reversal’s precondition, measured directly

Section 13 leaves an asymmetry. The extrapolation failure reproduced in full on Kleiber’s law; the ranking inversion did not, and the explanation offered there was a conjecture: with a single predictor a tree has little to exploit, never wins the interpolation split, and so has no advantage to lose. Two datasets cannot test that, because HDB5 and the mammalian compilation differ in feature count and also in science, group structure, sample size and noise.

A third dataset can, if the feature count is varied inside it while everything else is held fixed. The Biomass And Allometry Database [7] records, for individual plants, a nested set of measurements: basal diameter, height, leaf area and leaf mass. Taking the rows complete in all four gives 3599 plants across 53 species, 33 of them with the 30 individuals we require to hold a species out, spanning 36x in diameter across species medians and over seven orders of magnitude in mass. It is released CC0 and pinned by SHA-256 as the other deposits are. Species plays the part of tokamak, diameter the part of machine size, and the branching-network prediction that mass scales as diameter to the $8/3$ [8] the part of IPB98(y,2): an exponent derived from theory rather than fitted, which refits here to 2.512 and costs 0.5641 against

predictors	interp. gain	held-out gain	reversal
diameter	−1.7%	−10.5%	no
+ height	−0.6%	−10.8%	no
+ leaf area	+1.8%	−13.1%	yes
+ leaf mass	+6.8%	−13.4%	yes

Table 9: The best tree ensemble’s margin over the log-linear power law, on 3599 plants held fixed across the four rungs. Positive favours the trees. The interpolation column crosses zero; the extrapolation column does not, and grows slightly against them.

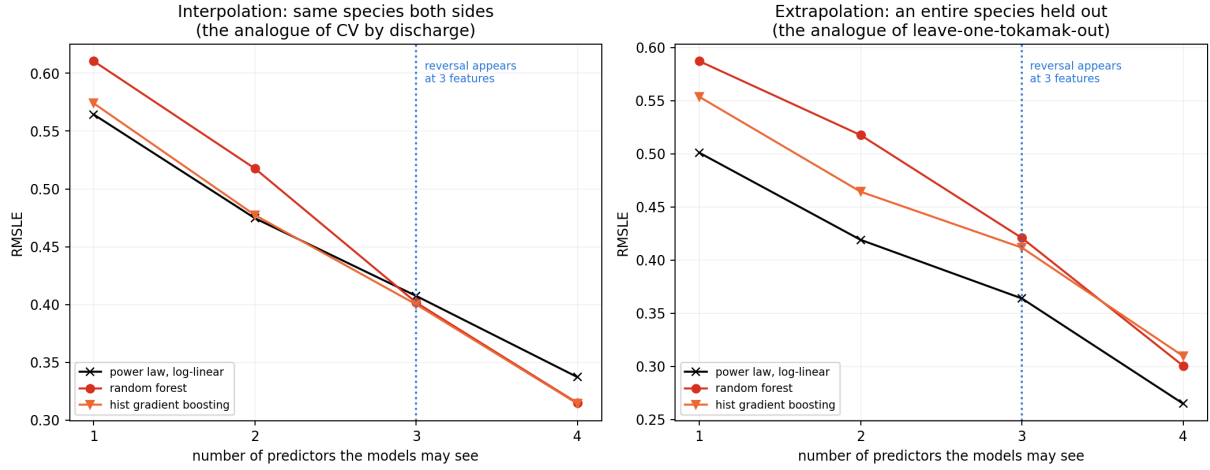


Figure 5: The reversal’s precondition, on 3599 plants held fixed. Left: the interpolation split, where every species appears on both sides. Right: an entire species held out. Only the number of predictors varies. The left curves cross; the right ones do not.

0.5780 in sample when held.

Each rung of the ladder is a prefix of the next and all four are scored on the same rows, so the only quantity that changes across them is how many predictors the models may see. The split structure matters and is easy to get wrong: grouped cross-validation by discharge keeps every machine on both sides, so its analogue keeps every species on both sides. Grouping the cross-validation by species instead would be a second leave-one-group-out, and against it no reversal can appear at any dimension.

Table 9 separates two statements that had been travelling as one. The interpolation margin rises monotonically and crosses zero between two and three predictors. The extrapolation margin does not shrink as the trees improve; it widens, and the power law wins the held-out species at 4 of 4 rungs. The inversion therefore appears at 3 predictors and every rung above, which is exactly where the interpolation margin turns positive.

So the extrapolation failure holds in tokamaks, mammals and trees alike, at every predictor count tested here. The *inversion* is conditional, and its condition has nothing to do with the failure: it is only whether the flexible model had enough to work with to win the comparison a practitioner runs first. The inversion is the warning light rather than the disease. Below three predictors the hazard is identical and the light is off.

15 Flexibility, or boundedness?

Section 5 attributes the failure to flexibility, and the polynomial ladder there supports that reading only for one kind of flexibility: polynomial degree under an isotropic penalty, swept over nine decades without rescue. Every model scored to this point confounds two properties. Flexibility is how much structure a model can learn beyond a power law. *Boundedness* is what it does far outside the training data. The tree ensembles here are flexible and bounded; ridge is neither; a degree-3 polynomial is flexible and unbounded but divergent, which is why its worst machine is 4.601. No model here was flexible, unbounded and well behaved at long range at once.

A Gaussian process [9] supplies one, and the property is a choice of kernel rather than a tuning

model	CV	held-out	ITER cut	ρ
GP, linear + RBF	0.112	0.218	0.191	-0.01
GP, RBF only	0.142	0.541	1.948	+0.65
GP, linear only	0.181	0.214	0.278	-0.06
random forest	0.128	0.465	0.938	+0.85
IPB98(y,2), not blind	0.199	0.188	0.194	-0.49
mean baseline	0.869	0.994	1.459	+0.56

Table 10: RMSLE under the three splits, and the rank correlation of per-machine error against extrapolation distance. Only the kernel’s long-range behaviour differs across the first three rows, and it spans a factor of ten at the ITER-matched cut. IPB98(y,2) was fitted with these machines included and is a reference point rather than a blind baseline.

parameter. We fit three over the same log features, with the same optimizer on the same rows, differing only in what the kernel does at long range. An RBF kernel decays to zero with distance, so the posterior returns to its prior mean, which centring places at the training mean: bounded. A dot-product kernel is Bayesian linear regression in the log features, which is a power law: unbounded and inflexible. Their sum is both. Hyperparameters are fitted by maximising the log marginal likelihood on a seeded subsample of each training fold, never on held-out rows; hand-chosen values are not adequate here, since a length scale an order of magnitude from the one the data supports makes the third rung look like a failure.

The control lands: the linear-kernel process and ridge agree to 0.0000 under cross-validation and on a held-out machine, and to 0.0007 at the ITER-matched cut, which is what licenses reading the other two rows as consequences of the kernel. The bounded rung then fails exactly as a tree does. It beats the power law under cross-validation and scores 1.948 at the ITER-matched cut, worse than predicting a constant, with error tracking distance at $\rho = +0.65$. The mechanism is the one in Sec. 5: a random forest cannot exceed the largest training target, and an RBF process reverts to its prior mean. Both are hard limits rather than shortfalls, and on the held-out rows this one’s predictions carry half the spread of the truth.

Adding a dot-product term to that same kernel changes nothing about its flexibility and everything about its behaviour at range. It gives the best cross-validated score in this work, 0.112 against the random forest’s 0.128; it scores 0.191 at the ITER-matched cut, better than the analytic law fitted with those machines included and second only to the constrained fit of Sec. 9 at 0.183; and its error is flat against distance at $\rho = -0.01$.

The consequence is a correction to this paper’s own diagnosis. The inversion is not a fact about flexible models but about bounded ones. The zoo of Sec. 5 is measured correctly and every flexible member of it was bounded or divergent, so nothing in it could separate the two properties. “Do not use a flexible model out of distribution” is the wrong lesson; “do not use a model whose predictions are bounded by its training targets” is the right one. This does not displace the constrained power law, which still wins the ITER-matched cut with nothing to tune, and the process does not beat the power law on a held-out machine (0.218 against 0.214).

A Gaussian process also supplies an interval by construction rather than by calibration, so Sec. 8’s question can be put to a model that was never calibrated. At the ITER-matched cut, at nominal 90%, the flexible unbounded rung covers 92.5%, where split-conformal intervals give the random forest 3% and the gradient booster none. It is the only model here whose out-of-distribution intervals reach nominal without being recalibrated to do so. The bounded rung is the instructive failure: it is the one model that does become vague out of distribution, with a half-width four times any other, and it still covers 16.7%, having reverted to a prior mean nowhere near the answer. Width was never the problem.

16 Limitations

The refit population is not IPB98’s, and the refit exponents should not be read as a correction to it. Two machines (JET and ASDEX Upgrade with their variants) supply 77% of rows. Only 13 of 18 machines are scored under leave-one-out; the five excluded are the smallest and most unlike the rest, so the reported gap is if anything optimistic. Thirteen machines is a small bootstrap and the paired differences are the more reliable statistic. The training-target bound of Sec. 5.1 is a theorem for the random forest and an observation for the gradient booster: boosting sums tree outputs rather than averaging targets, so nothing confines it to the training range, and what is reported here is only that it stayed inside that range on all 14 splits scored. The ITER-matched cut reproduces ITER’s size ratio and nothing else: ITER is a burning plasma dominated by alpha self-heating at a gain no row here approaches, so clearing this bar

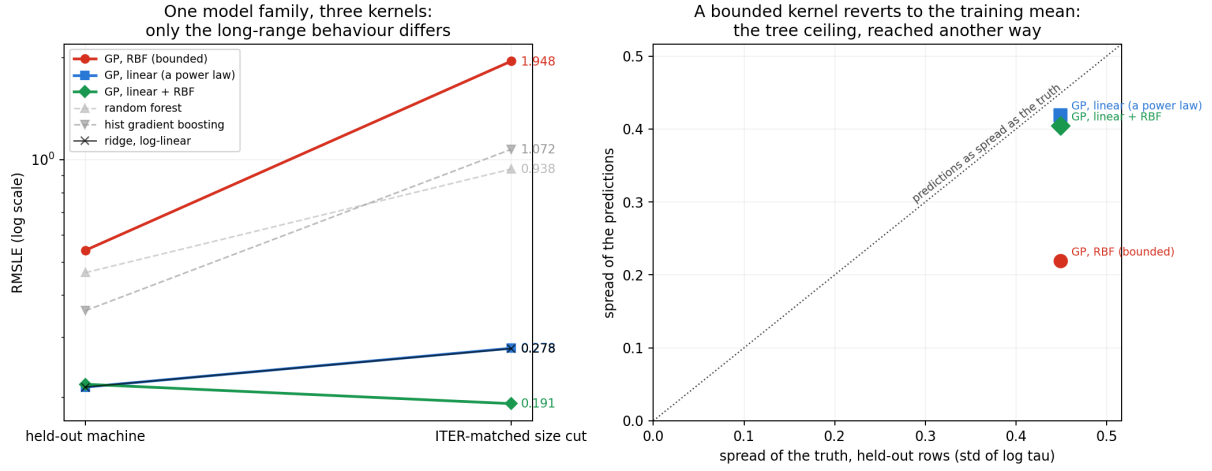


Figure 6: One model family, three kernels. Left: RMSLE on a held-out machine and at the ITER-matched cut; only the kernel’s long-range behaviour differs. Right: spread of the predictions against spread of the truth on the held-out rows. A model that has reverted to its prior mean sits near zero however good its in-range fit was.

is necessary and not sufficient. The per-machine coverages in Sec. 8 are proportions over as few as 39 rows and carry binomial uncertainty of ten points or more; the pooled contrasts are far larger than that noise, the within-column ordering is not. The constraint hierarchy of Sec. 9 has an optimum in its middle that nothing here predicted and that cross-validation cannot select, so the result is that a constraint helps rather than that this constraint is the right one. The replication of Sec. 11 is disjoint in rows but not independent in provenance: both arms come from the same ITPA collection and the same devices, and its five-machine arm is too small to carry a claim alone. The locked forecast of Sec. 12 is a record, not a validation; only JT-60SA can be checked against measurement in the near term, that check requires the H-mode campaign beginning in late 2026 rather than the completed L-mode one, and it is the one device of the three that sits inside the database’s size range. The replication of Sec. 13 has one predictor and eleven groups, and those groups are phylogenetically related rather than independent, which the comparative-biology literature handles with phylogenetically controlled regression [21] and we do not; it suffices to show the extrapolation failure is not fusion-specific and that the reversal needs a condition that dataset lacks, not to establish a rate for either. The ladder of Sec. 14 is one ordering of one set of predictors, so it establishes that a crossing exists and that the inversion tracks it, not that three is a threshold; its rows are a selective intersection, cutting 21084 plants to 3599 and favouring studies that measured leaves destructively, most of those plants are glasshouse-grown or from common gardens rather than wild forest, and its 14 source studies mean holding out a species sometimes holds out a laboratory. The kernels of Sec. 15 are tuned by marginal likelihood on a seeded subsample of each training fold rather than on all of its rows, which is stable across the range we could afford (under 4% in every hyperparameter between 750 and 1500 rows) but is not a proof that the full-data optimum agrees; and “linear plus RBF” is one reading of “a power law with a bounded correction” among several, with an anisotropic kernel, a Matern kernel and a linear part constrained as in Sec. 9 all untested. The last of these is the obvious next experiment, since the constraint and the kernel are the two things that work here and nothing has tried them together.

17 Conclusion

On real confinement data, the model that wins cross-validation by 36% loses to a 1998 power law on every machine it has not seen, and the failure has three measurable mechanisms rather than a tuning fix. At the size extrapolation that separates this database from ITER, the best cross-validated models land closer to a constant predictor than to the power law, and their nominal 90% intervals cover 3% and 0% of the held-out rows at unchanged width. A power law is used in this field not because it fits better, since it fits worse, but because it is the functional form that survives leaving the data behind. A bounded correction on its residuals improves it in the one direction a next-step device lies in, and the same boundedness that disqualifies trees as predictors is what makes that correction safe.

The strongest repair, though, is not a model at all. Constraining the exponents to a surface the physics requires them to lie on beats every model built here at the ITER-matched cut, costs 0.001 in sample, and has nothing to tune. Supplying the same physics as a prior is worth substantially less: a constraint names

a surface where a prior names a point, and the weaker information is the more useful one. The reversal itself reproduces on rows the standard set does not contain and in a second confinement regime, so it is a property of how these models are validated rather than of one file. What the models disagree about is recorded rather than argued: on a machine inside the database’s size range they agree to within 15%, and on ITER they differ by a factor of 8.3.

Two results temper the diagnosis and neither weakens the practical advice. Varying only the number of predictors shows that the extrapolation failure persists at every predictor count tested while the ranking inversion does not: the inversion needs the flexible model to win the interpolation split first, so a field whose scaling law has one predictor faces the same danger with no warning at all. And separating flexibility from boundedness shows the failure belongs to the second. A Gaussian process with a bounded kernel fails as a tree does, while the same flexibility carried on an unbounded one gives the best cross-validated score here, holds up at the ITER-matched cut, and is the only model in this work whose out-of-distribution intervals reach nominal without recalibration. The lesson is not that flexible models cannot be trusted beyond their data. It is that a model whose long-range predictions are bounded by the range of its training targets is structurally unsuited to a problem where the target itself has to leave that range, and that the two are easy to confuse because every flexible model in the standard zoo is also bounded.

Data and code availability

The HDB5 STD5 dataset is third-party data obtained from OSF (<https://osf.io/drwcq>) and is not redistributed here; it is pinned by SHA-256 and verified on load. All code, the full writeup with every table and limitation, and the scripts that regenerate every number and figure in this paper are available in the accompanying repository.

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